ACADEMIC PROJECT

CosmoE

An AI-Powered Product Search and Recommendation System for Home Decor

The complete project, explained in simple words

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Built with Python, Django, BERT and spaCy

1. What is CosmoE?

CosmoE is a website that helps people find **home decor, kitchen and wooden products**. Instead of visiting many shopping websites one by one, a user can come to CosmoE and see products from **Daraz.pk** and **Etsy** in one place.

What makes CosmoE special is its **search**. Most shopping websites only look for the exact words you type. CosmoE uses **Artificial Intelligence (AI)** to understand *what you mean*, so it can find the right product even when you describe it in your own words.

> **In one sentence**
> CosmoE collects products from online shops, saves them, and uses AI to show you the products that best match what you are looking for.

Think of it like this

Imagine a helpful shopkeeper. You say, *"I want something wooden to put on my wall."* A normal search box would only look for products containing those exact words. A good shopkeeper understands you and shows you wall shelves, wall hangings and wooden wall art. **CosmoE tries to be that good shopkeeper.**

2. The problem it solves

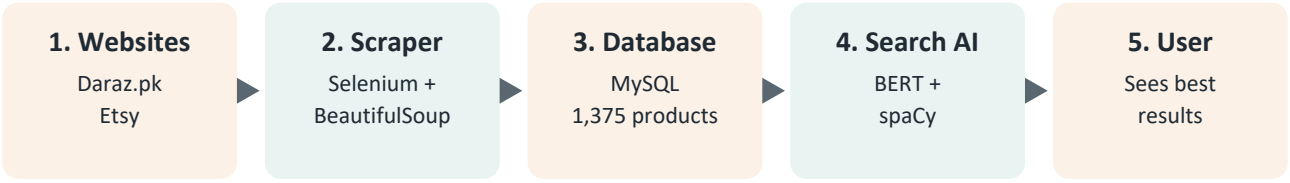
Online shoppers face three common problems:

- **Products are scattered.** The same type of product is sold on many websites, so people have to open and compare several sites themselves.
- **Search is too literal.** If you search *"wooden wall art"*, a normal search may miss a product called *"Laser Cut Wall Hanging for Living Room"*, even though it is exactly what you want.
- **Too much information.** Product titles are very long and full of extra words, so it is hard to quickly find the best, highest-rated items.

> **The question this project answers**
> Can AI language models help people find products by understanding the **meaning** of their search, and not only the exact words?

3. How it works: the big picture

The whole project can be understood as five simple steps. Products move from the shopping websites, through CosmoE, and finally to the user.



Products travel from left to right: collected, stored, searched, then shown.

Step	What happens	In simple words
1. Websites	Daraz.pk and Etsy list thousands of products.	Where the products come from.
2. Scraper	A program opens the websites and copies product details.	A robot that reads web pages for us.
3. Database	The details are saved in a MySQL database.	A big organised notebook.
4. Search AI	BERT and spaCy work out which products match a search.	The "brain" that understands your words.
5. User	The website shows the best matching products.	You see the results.

4. Step 1: Collecting products (web scraping)

Web scraping means using a computer program to read a website and copy information from it automatically, just as a person would copy details by hand, but much faster.

CosmoE uses two tools for this:

- **Selenium** opens a real Google Chrome browser and loads the shopping pages, just like a person visiting the site. This matters because many modern websites only show products after the page has fully loaded.
- **BeautifulSoup** reads the page's HTML code and picks out the useful parts: the product name, price, rating, number of reviews, number sold, picture and link.

Every website is built differently, so the scraper has separate instructions for Daraz and for Etsy. It also handles small problems. For example, when Daraz shows a blank placeholder picture, the scraper opens that product's own page to fetch the real picture.

> **Example of what one collected product looks like**
>
> **Title:** Wooden Wall Hanging, Laser Cut Beautiful Style Design for Home Decor
> **Rating:** 4.2/5 **Reviews:** 120 **Sold:** 1K **Price:** Rs. 299
> **Source:** Daraz.pk

5. Step 2: Storing products (database)

After collecting the products, CosmoE saves them in a **MySQL database**. A database is like a spreadsheet that a program can read and search very quickly. Each product is saved as one row, with the following columns:

Column	What it stores
title	The product's name
rating	Customer rating, for example 4.3/5
review_count	How many customers left a review
sold_count	How many units were sold
price	The product's price
image_src	Link to the product picture
addtocart_href	Link to buy the product on the original website
created_at	The date and time the product was collected

CosmoE uses **Django**, a popular Python web framework. Django lets the project describe this table in a few lines of Python code (called a **model**), and it creates the real database table automatically. Django also provides a ready-made **admin panel** where the project owner can view, search and filter all saved products.

The products are also exported to a **CSV file** (a simple spreadsheet file) so the AI part of the project can use them.

6. Step 3: Smart Search with BERT

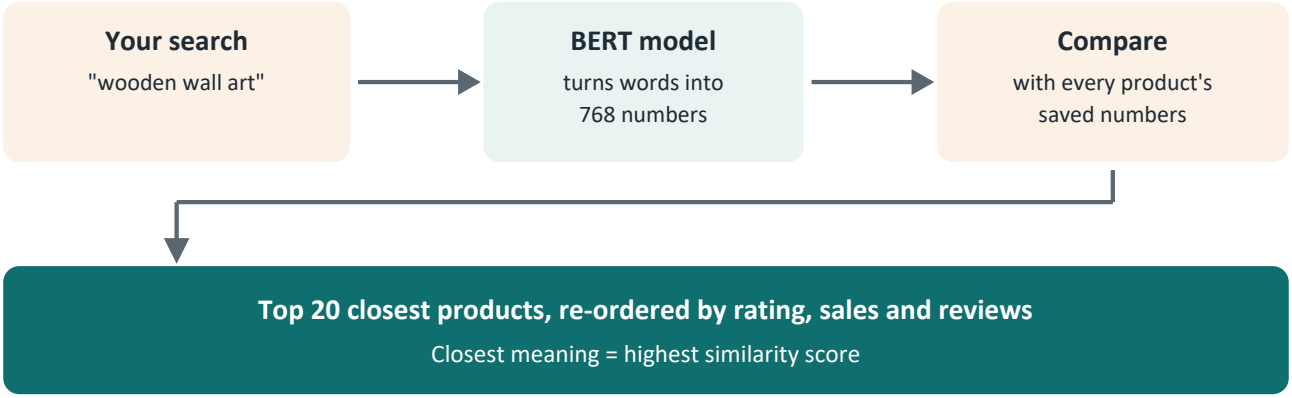
This is the most important and most advanced part of the project.

What is BERT?

BERT is an AI language model created by Google. It has read a huge amount of English text, so it has learned how words relate to each other. It knows, for example, that *"wall hanging"* and *"wall art"* mean similar things.

How CosmoE uses BERT

Computers cannot compare meanings directly, but they are very good at comparing numbers. So BERT turns every piece of text into a list of **768 numbers**, called an **embedding**. You can think of an embedding as a **"meaning fingerprint"**: texts with similar meanings get similar fingerprints.



- **Done once, in advance:** every product title is turned into its fingerprint and saved in a file (*product_embeddings.pkl*). This makes searching fast, because the products don't need to be processed again on every search.
- **Done on every search:** the user's search text is turned into a fingerprint using the same BERT model.
- **Compare:** a formula called **cosine similarity** measures how close the search fingerprint is to each product fingerprint. A higher score means the meanings are more similar.
- **Pick and sort:** the 20 closest products are chosen, then sorted by **rating**, **number sold** and **number of reviews**, so good-quality products appear first.

> **Why this is better**
>
> Because BERT compares meanings, a search for *"wooden wall art"* can also find products described as *"wall hanging"* or *"wall decoration"*, even without the exact same words.

7. Step 4: Keyword Search with spaCy

CosmoE also has a second, simpler search called **Formal Search**. It uses **spaCy**, a well-known Python library for working with language (Natural Language Processing, or NLP).

How it works

- **Clean the search:** spaCy removes common filler words like *"the", "for", "a", "with"* (called **stop words**) and punctuation marks.
- **Keep the important words:** only the meaningful keywords are left.
- **Score each product:** CosmoE counts how many of those keywords appear in each product title.
- **Show the best:** products with the most matching keywords are shown first (up to 30 results).

Example

```
User types:      "a beautiful wooden shelf for the kitchen"
After cleaning:  beautiful, wooden, shelf, kitchen

Product A title contains: wooden, shelf, kitchen   -> score 3  (shown first)
Product B title contains: wooden, shelf            -> score 2
Product C title contains: kitchen                  -> score 1
```

Smart Search vs. Keyword Search

	Smart Search (BERT)	Keyword Search (spaCy)
Looks at	The meaning of the words	The exact important words
Finds similar words?	Yes	No
Speed	Slower (runs a large AI model)	Fast
Results shown	Top 20	Top 30
Best for	Describing what you want in your own words	Searching for specific words

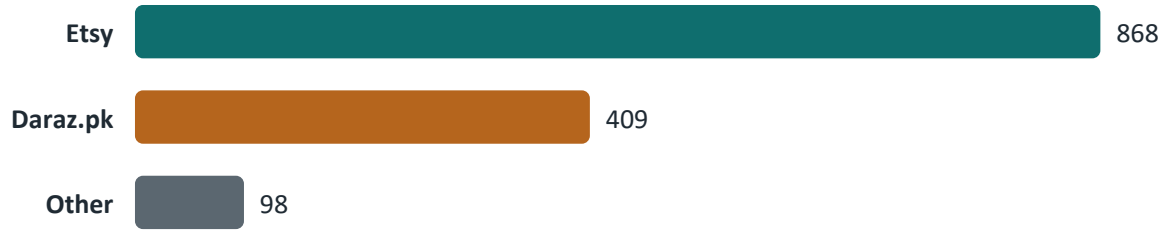
8. The website pages

The website is built with HTML, CSS and jQuery, and it has the following pages:

Page	Web address	What it shows
Home	/	Products sorted by rating, 20 per page
Smart Search	/search/	AI search results using BERT
Formal Search	/formalsearch/	Keyword search results using spaCy
Category	/category/<name>/	Home Decor, Kitchen or Wooden products
Scraper	/scrap/	Starts collecting new products
About Us	/about-us/	Information about the project
Contact Us	/contact-us/	Contact information
Admin	/admin/	Control panel for managing products

9. The data we collected

The project's own scraper collected **1,375 real products**. Here is where they came from:



Each product includes its name, price, rating, reviews, sales count, picture and a link to the original shop. The full dataset is included in the project as a CSV file (*csv_files/scrapdata_productrecord.csv*).

10. Tools and technologies

Tool	What it is	Why the project uses it
Python	Programming language	The whole project is written in Python
Django	Web framework	Builds the website, pages and admin panel
MySQL	Database	Stores all the collected products
Selenium	Browser automation	Opens shopping websites automatically
BeautifulSoup	HTML reader	Picks product details out of web pages
BERT (Transformers)	AI language model	Understands the meaning of searches
PyTorch	Deep learning library	Runs the BERT model
spaCy	NLP library	Finds the important words in a search
scikit-learn	Machine learning library	Calculates cosine similarity
pandas / NumPy	Data libraries	Handle the product data and number lists
HTML, CSS, jQuery	Web technologies	Design and layout of the website

11. How to run the project

You need **Python 3.11**, **MySQL**, and **Google Chrome** (Chrome is only needed for the scraper). Then follow these steps:

Step 1: Install the required libraries

```
pip install django==4.2.5 mysqlclient algoliasearch-django selenium
pip install webdriver-manager beautifulsoup4 transformers torch
pip install spacy scikit-learn pandas numpy
python -m spacy download en_core_web_lg
```

Step 2: Create the database

In MySQL, create an empty database called **cosmoe**. Then let Django create the tables:

```
python manage.py migrate
```

Step 3: Add products

Either import the included CSV file into the database, or start the website and open */scrap/* to collect fresh products.

Step 4: Start the website

```
python manage.py runserver
```

Then open **http://127.0.0.1:8000/** in your web browser.

12. Limitations and future work

What could be better today

- **New products need extra work:** after scraping new products, their meaning fingerprints must be recalculated before Smart Search can find them.
- **Websites change:** if Daraz or Etsy redesign their pages, the scraper needs updating.
- **Only titles are used:** search looks only at product names, not pictures or full descriptions.
- **General AI model:** BERT was trained on general English text, not specifically on shopping data.

Ideas for the future

- Train the AI model on shopping data (or use Sentence-BERT) for more accurate results.
- Search using pictures as well as text (for example with the CLIP model).
- Give personal recommendations based on what each user likes.
- Collect new products automatically on a schedule.
- Measure search quality with standard scores such as Precision@K and NDCG.
- Put the website online for public use.

13. Glossary of terms

Simple explanations of the technical words used in this guide:

Term	Simple meaning
Artificial Intelligence (AI)	Computer programs that can do tasks that normally need human thinking.
NLP	Natural Language Processing: teaching computers to understand human language.
Web scraping	Using a program to automatically copy information from websites.
Database	An organised store of information that programs can search quickly.
Django	A Python toolkit for building websites.
BERT	An AI model by Google that understands the meaning of English sentences.
Embedding	A list of numbers that represents the meaning of a piece of text (a "meaning fingerprint").
Cosine similarity	A formula that measures how similar two embeddings are. Higher means more similar.
Stop words	Very common words like "the" or "for" that carry little meaning.
Token	A small piece of text, such as a word or part of a word, that the AI reads.
CSV file	A simple spreadsheet file where values are separated by commas.
Admin panel	A private control page where the owner manages the website's data.

14. Summary

CosmoE is a complete project that goes all the way from **collecting real data** to a **working, AI-powered website**:

- It **collects** 1,375 real products from Daraz.pk and Etsy using a web scraper.
- It **stores** them in a MySQL database managed by Django.
- It **understands** searches using the BERT AI model and cosine similarity.
- It **also offers** a fast keyword search built with spaCy.
- It **shows** the best results first, ranked by rating, sales and reviews.

> **What this project demonstrates**
>
> Practical skills in web development, data collection, databases, Natural Language Processing and deep learning, brought together to solve a real-world problem: helping people find the products they want, more easily.